

Marking Scheme

2026 · Paper Paper 1 · 105 marks

General Rules

- M / A / PROOF mark types
- **f.t.** : M marks always follow through. A marks follow through ONLY when isFollowThrough = true.
- Deduction cap : $A(1)=1 \cdot A(2)=1 \cdot B=0$ (u, pp)
- No deduction in any step scoring 0.
- A correct answer merits all marks unless a method is specified.
- Different notation is NOT penalized. · Benefit of doubt
- All fractional answers must be simplified.

Section A(1) (35 marks)

1. Laws of Indices [B-]

a (3 marks)

M1 $(u^{-3}v)^2 = u^{-6}v^2$
for expanding the power / 展開次方

M1 $\frac{u^9v^{-4}}{u^{-6}v^2} = u^{9-(-6)}v^{-4-2} = u^{15}v^{-6}$ **f.t.**
subtract indices / 相除時指數相減

A1 $= \frac{u^{15}}{v^6}$
positive indices required / 必須以正指數表示

2. Change of Subject [B-]

a (3 marks)

M1 $5h + 2 = k(h - 4)$
cross multiplication / 交叉相乘

M1 $5h - kh = -4k - 2 \Rightarrow h(5 - k) = -(4k + 2)$ **f.t.**
group h terms / 集合含 h 的項

A1 $h = \frac{4k + 2}{k - 5}$
or equivalent

3. Estimation and Error [B-]

a (1 mark)

A1 $2400 - \frac{100}{2} = 2350 \text{ g}$
u-1 if unit g missing / 漏寫單位 g 扣 1 分

b (2 marks)

M1 **Greatest possible total = $6 \times 2450 = 14700 \text{ g}$** **f.t.**
use upper limit 2450 g / 使用上限 2450 g

A1 **$14650 < 14700$, *so it IS possible*** **f.t.**
1A f.t. — conclusion required / 結論句必須寫出

4. Factorization [B]

a (2 marks)

M1 $25x^2 - 30xy + 9y^2 = (5x)^2 - 2(5x)(3y) + (3y)^2$
recognize perfect square / 辨認完全平方式

A1 $= (5x - 3y)^2$

b (2 marks)

M1 $= (5x - 3y)^2 + 2(5x - 3y)$ f.t.
use result of (a) / 引用 (a)

A1 $= (5x - 3y)(5x - 3y + 2)$
pp-1 if not fully factorized / 未完全分解扣 1 分

5. Percentages [B]

a (4 marks)

M1 $Let \text{ } C \text{ be the cost. } 32$
set up profit equation / 建立利潤方程

A1 $C = 700$

M1 $Selling\ price = 700 + 224 = 924; MP \times (1 - 30$ f.t.
SP = MP x 70% / 售價 = 標價 x 70%

A1 $Marked\ price = 1320$
u-1 if missing / 漏寫扣 1 分

6. Linear Inequalities [B]

a (3 marks)

M1 $4(x + 2) \geq x - 1 \Rightarrow 3x \geq -9 \Rightarrow x \geq -3$

M1 $7 - 2x > x - 8 \Rightarrow 15 > 3x \Rightarrow x < 5$

A1 $Taking\ the\ intersection : -3 \leq x < 5$ f.t.
'and' -> intersection / 「且」取交集

b (1 mark)

A1 -3 f.t.
1A f.t. from (a)

7. Transformation in the Rectangular Coordinate Plane [B+]

a (2 marks)

A1 $A' = (-8, 6)$
reflection about the line $y = x$

A1 $B' = (5, 3)$
clockwise 90 degrees

b (2 marks)

M1 $slope = \frac{3 - 6}{5 - (-8)}$ f.t.
1M f.t. from (a)

A1 $= -\frac{3}{13}$

8. Congruent Triangles [B+]

a (2 marks)

PROOF $AB = AD(\text{given}); \angle BAC = \angle DAC(\text{given}); AC = AC(\text{common})$

Case 1 with reasons = 2; Case 2 without reasons = 1

PROOF $\therefore \triangle ABC \cong \triangle ADC(\text{SAS})$

reason SAS required / 須寫理由「邊、角、邊」

b (3 marks)

M1 $\angle ADC = \angle ABC = 100^\circ(\text{corr. } \backslash \text{angles, } \cong \backslash \text{triangles})$ **f.t.**

use result of (a) / 引用(a)

M1 $\angle BAD + \angle ABC + \angle BCD + \angle ADC = 360^\circ$

angle sum of quadrilateral / 四邊形內角和

A1 $\angle BAD = 360^\circ - 100^\circ - 100^\circ - 96^\circ = 64^\circ$

u-1 if degree symbol missing / 漏寫「度」扣1分

9. Statistical Tables and Probability [B+]

a (1 mark)

A1 $16 - 20\text{minutes}$

b (2 marks)

M1 $\text{Mean} = \frac{6(8) + 12(13) + 20(18) + 14(23) + 8(28)}{60} = \frac{1110}{60}$

use class marks / 使用組中點

A1 $= 18.5\text{minutes}$

u-1 if unit missing / 漏寫單位扣1分

c (2 marks)

M1 $P = \frac{6 + 12}{60}$

A1 $= \frac{3}{10}$

simplify fraction / 分數須化至最簡

Section A(2) (35 marks)

10. Variation [A-]

a (3 marks)

M1 $f(x) = a + bx^2, b \neq 0$

correct form / 部分變分形式

M1 $a + b = 9, a + 9b = 41 \Rightarrow 8b = 32$ **f.t.**

form simultaneous equations / 建立聯立方程

A1 $b = 4, a = 5 \Rightarrow f(x) = 5 + 4x^2$

both correct / 兩者皆正確

b (1 mark)

A1 5 **f.t.**

1A f.t. from (a)

c (3 marks)

M1 $5 + 4x^2 = k \Rightarrow x^2 = \frac{k-5}{4}$ **f.t.**

M1 $\text{Two distinct real roots} \Leftrightarrow \frac{k-5}{4} > 0$ **f.t.** **alt: q10c_alt**

or use discriminant / 或用判別式

A1 $k > 5$

11. Measures of Central Tendency and Dispersion [A-]

a (3 marks)

M1 **Median = the 6th datum = $20 + a = 26 \Rightarrow a = 6$**
median = 6th of 11 data / 中位數為 11 個數據的第 6 個

M1 **$Q_1 = 19, Q_3 = 30 + b, Q_3 - Q_1 = 17 \Rightarrow b = 6$** ft.
 $Q_1 = 3rd\ datum, Q_3 = 9th\ datum$

A1 **$a = 6, b = 6$**
both correct / 兩者皆正確

b (2 marks)

A1 **Mean = $\frac{310}{11} \approx 28.2$** ft.
r.t. 28.2

A1 **Standard deviation ≈ 10.6** ft.
*1A r.t. 10.6
r.t. 10.6*

c (2 marks)

M1 **New median = $\frac{24 + 26}{2} = 25$** ft.

A1 **The median decreases by $26 - 25 = 1$** ft.
1A f.t. — direction must be stated / 須說明增減方向

12. Remainder Theorem and Factor Theorem [A]

a (3 marks)

M1 **$p(x) = (x^2 - 2x + 3)(2x + t) + (-5x - 15)$**
division algorithm / 除法算式

M1 **Constant : $3t - 15 = -18 \Rightarrow t = -1$** ft.
compare coefficients / 比較係數

A1 **$a = t - 4 = -5, b = 3$**
both correct / 兩者皆正確

b (2 marks)

M1 **$p(3) = 2(27) - 5(9) + 3(3) - 18$** ft.
factor theorem / 因式定理

A1 **$p(3) = 0 \Rightarrow x - 3 \text{ is a factor}$** ft.
1A f.t. — conclusion required / 結論句必須寫出

c (2 marks)

M1 **$p(x) = (x - 3)(2x^2 + x + 6); \Delta = 1^2 - 4(2)(6) = -47 < 0$** ft.
1M for factorization OR discriminant

A1 **$\Delta < 0 \Rightarrow \text{only ONE real root } x = 3, \text{ so DISAGREE}$** ft.
1A f.t. — conclusion required / 結論句必須寫出

13. Volumes and Surface Areas of Similar Solids [A]

a (3 marks)

M1

$$\text{Volume of larger cone} = \frac{1}{3}\pi(15)^2(20) = 1500\pi \text{ cm}^3$$

cone volume formula / 圓錐體積公式

M1

$$\frac{V_{\text{small}}}{1500\pi} = \frac{64}{125} \quad \text{ft.}$$

use volume ratio / 運用體積比

A1

$$V_{\text{small}} = 768\pi \text{ cm}^3$$

u-1 if cm³ missing / 漏寫 cm³ 扣 1 分

b (2 marks)

M1

$$\pi(6)^2 h = 1500\pi + 768\pi \quad \text{ft.}$$

conservation of volume / 體積守恒

A1

$$h = \frac{2268}{36} = 63 \text{ cm}$$

u-1 if cm missing / 漏寫 cm 扣 1 分

c (2 marks)

M1

$$\text{Curved SA of cylinder} = 2\pi(6)(63) = 756\pi; \text{slant height} = \sqrt{15^2 + 20^2} = 25 \quad \text{ft.}$$

1M f.t. from (b)

A1

$$\pi(15)(25) = 375\pi; 756\pi > 375\pi, \text{so AGREE} \quad \text{ft.}$$

1A f.t. — conclusion required / 結論句必須寫出

14. Equations of Circles [A]

a (2 marks)

A1

$$G = (6, 10)$$

A1

$$r = \sqrt{6^2 + 10^2 - 72} = \sqrt{64} = 8$$

b (1 mark)

A1

$$GH = \sqrt{(21 - 6)^2 + (10 - 10)^2} = 15 \quad \text{ft.}$$

c (1 mark)

A1

$$GP \perp GH$$

accept 'GP is perpendicular to GH' / 亦接受「GP 垂直於 GH」

d (3 marks)

M1

$$GP = r = 8, GH = 15, \angle HGP = 90^\circ \quad \text{ft.}$$

M1

$$HP = \sqrt{8^2 + 15^2} = 17 \quad \text{ft.}$$

Pythagoras / 畢氏定理

A1

$$\text{Perimeter} = 8 + 15 + 17 = 40$$

Section B (35 marks)

15. Permutations, Combinations and Probability [A]

a (3 marks)

M1 **Total** = $C_4^{15} = 1365$

M1 **Favourable** = $C_2^7 \times C_2^8 = 21 \times 28 = 588$ ft.

A1 $P = \frac{588}{1365} = \frac{28}{65}$
simplify fraction / 分數須化至最簡

b (3 marks)

M1 **P** = $1 - P(\text{no red}) - P(\text{no green}) + P(\text{no red and no green})$
inclusion-exclusion / 容斥原理

M1 $= \frac{1365 - C_4^8 - C_4^{12} + C_4^5}{1365} = \frac{1365 - 70 - 495 + 5}{1365}$ ft.

A1 $P = \frac{805}{1365} = \frac{23}{39}$
1A r.t. 0.590
r.t. 0.590

16. Graphs of Quadratic Functions [A]

a (2 marks)

M1 **f(x)** = $2(x - 2k)^2 - 8k^2 + 10k^2 + 24k + 7$
completing the square / 配方法

A1 **Vertex** = $(2k, 2k^2 + 24k + 7)$

b (2 marks)

M1 $2k^2 + 24k + 7 = 2(k + 6)^2 - 65$ ft.
complete the square again / 再次配方

A1 **Least value** = -65 when $k = -6$
both correct / 兩者皆正確

c (2 marks)

M1 **Put x = 2k into Γ** : $y = \frac{1}{2}(2k)^2 + 12(2k) + 7 = 2k^2 + 24k + 7$ ft.
1M f.t. from (a)

A1 **Equal to they – coordinate of the vertex, so AGREE** ft.
1A f.t. – conclusion required / 結論句必須寫出

17. Arithmetic and Geometric Sequences [A]

a (2 marks)

M1 **d** = $\frac{41 - 11}{7 - 2} = 6$
find common difference / 求公差

A1 **A(n)** = $11 + (n - 2)(6) = 6n - 1$

b (1 mark)

A1 **S(40)** = $\frac{40}{2}[5 + 239] = 4880$ ft.

c (3 marks)

M1 **G(1) = A(3) = 17** $\Rightarrow \sum_{i=1}^n G(i) = 17(2^n - 1)$ ft.
sum of geometric sequence / 等比求和

M1 $17(2^n - 1) > 4880 \Rightarrow 2^n > 288.1 \Rightarrow n > \log_2 288.1 \approx 8.17$ ft.
accept trial with n = 8, 9 / 亦接受試值

A1 **Least n** = 9 : $8687 > 4880$; $n = 8$: $4335 < 4880$

18. Trigonometry in 3-Dimensional Problems [A+]

a (2 marks)

M1 $OM = 8; VM^2 = VO^2 + OM^2 = 15^2 + 8^2$

Pythagoras in triangle VOM

A1 $VM = \sqrt{289} = 17 \text{ cm}$

b (2 marks)

M1 $\text{Required angle} = \angle VMO; \tan \angle VMO = \frac{15}{8}$ **ft.**

1M identify correct angle / 正確辨認二面角

A1 $\angle VMO \approx 61.9^\circ$

1A r.t. 61.9 degrees

r.t. 61.9

c (4 marks)

M1 $VM \perp BC \text{ and } VN \perp AD; BC \parallel AD \Rightarrow \text{required angle} = \angle MVN$

1M identify angle between planes / 辨認兩平面夾角

M1 $VM = VN = 17 \text{ cm}, MN = 16 \text{ cm}$ **ft.**

M1 $\cos \angle MVN = \frac{17^2 + 17^2 - 16^2}{2(17)(17)} = \frac{322}{578}$ **ft.**

cosine formula / 餘弦公式

A1 $\angle MVN \approx 56.1^\circ < 60^\circ, \text{ so AGREE}$ **ft.**

1A f.t. r.t. 56.1 degrees

r.t. 56.1

19. Equations of Circles and Tangents [A+]

a (2 marks)

A1 $G = (5, 12)$

A1 $r = \sqrt{5^2 + 12^2 - 0} = 13$

b (3 marks)

M1 $\text{Distance from } G \text{ to } L = \frac{|5(5) + 12(12) - 338|}{\sqrt{5^2 + 12^2}} = \frac{169}{13} = 13$

point-to-line distance / 點到直線距離

PROOF $\text{Distance} = 13 = r \Rightarrow L \text{ is a tangent to } C$ **ft.**

conclusion required / 必須寫結論

A1 $T = (10, 24)$

T = foot of perpendicular from G to L / T 為 G 至 L 的垂足

c (2 marks)

M1 $O = (0, 0) \text{ satisfies } C; \text{ mid-point of } OT = (5, 12) = G$ **ft.** **alt: q19c_alt**

or show OT = 26 = 2r / 或證 OT = 26 = 2r

A1 $\text{Mid-point of } OT \text{ is the centre } G, \text{ so AGREE}$ **ft.**

1A f.t. — conclusion required

d (2 marks)

M1 $\text{Taking } OT \text{ as base, greatest height} = r = 13$ **ft.**

A1 $S_{\max} = \frac{1}{2}(26)(13) = 169$